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SEMINAR REPORT

ON

PHISHING WEBSITE DETECTION USING MULTI-INPUT CNN

Submitted by

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CERTIFICATE

This is to certify that Mr. Kashin Bhardwaj of T.Y. B.Tech., School of Computer Engineering & Technology, Department of Computer Engineering & Technology, CSE-Core/ CSE-AIDS/ CSE-CSF, Semester – VI, PRN. No. 1032230861, has successfully completed seminar on

Phishing Website Detection using multi-input CNN model

To my satisfaction and submitted the same during the academic year 2025 - 2026 towards the partial fulfillment of degree of Bachelor of Technology in School of Computer Engineering & Technology under Dr. Vishwanath Karad MIT- World Peace University, Pune.

Seminar Guide

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ABBREVIATIONS

CNN – Convolutional Neural Network

URL – Uniform Resource Locator

HTML – HyperText Markup Language

DOM – Document Object Model

ML – Machine Learning

DL – Deep Learning

LSTM – Long Short-Term Memory

RF – Random Forest

API – Application Programming Interface

GPU – Graphics Processing Unit

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ABSTRACT

Phishing attacks continue to be one of the most significant threats in the field of cybersecurity, targeting users by imitating trusted websites and services to obtain sensitive information such as login credentials, financial data, and personal details. With the rapid increase in online transactions and digital platforms, detecting such malicious activities has become more challenging, especially in the case of newly created or zero-day phishing websites.

This seminar presents a deep learning-based approach for phishing website detection using a multi-input Convolutional Neural Network (CNN) model. Unlike traditional methods that rely on a single type of input, the proposed system analyzes multiple aspects of a webpage, including URL patterns, HTML content, and visual appearance through screenshots. A character-level CNN is used to extract meaningful patterns from textual data such as URLs and HTML, while a separate CNN processes visual features from webpage images.

The extracted features from both models are combined using a feature fusion technique, allowing the system to make more accurate and reliable predictions. This approach improves the ability to detect sophisticated phishing attacks that may bypass conventional detection systems. The proposed model aims to provide a balance between detection accuracy and computational efficiency, making it suitable for real-world applications such as browser extensions and online security systems.

Keywords:

Phishing Detection, Convolutional Neural Network, Multi-Input Model, URL Analysis, HTML Feature Extraction, Visual Similarity Detection

1. INTRODUCTION

1.1 Background

With the rapid growth of the internet and digital services, users increasingly rely on online platforms for banking, communication, shopping, and information exchange. While this has improved convenience and accessibility, it has also created opportunities for cybercriminals to exploit unsuspecting users. One of the most common and dangerous forms of such cyber threats is phishing.

Phishing is a technique in which attackers create fraudulent websites or messages that appear to be legitimate in order to deceive users into revealing sensitive information. These websites are often designed to closely resemble trusted platforms, making it difficult for users to distinguish between genuine and malicious content. As a result, phishing attacks can lead to financial loss, identity theft, and unauthorized access to personal accounts.

1.2 Problem Statement

Traditional phishing detection methods rely heavily on blacklist-based systems and predefined rules. Blacklists maintain a database of known malicious URLs, but they are ineffective against newly created phishing websites, also known as zero-day attacks. Similarly, rule-based systems depend on manually crafted features such as URL length or the presence of suspicious keywords. These methods are limited in their ability to adapt to evolving attack patterns.

Another limitation of existing approaches is that they often focus on a single type of data. For example, some models analyze only the URL structure, while others focus on HTML content or visual features independently. This lack of comprehensive analysis reduces the overall effectiveness of phishing detection systems.

1.3 Need for the Study

As phishing techniques become more sophisticated, there is a need for intelligent systems that can automatically learn complex patterns from multiple sources of information. A detection

system that combines different types of data—such as textual and visual features—can provide a more complete understanding of a webpage and improve detection accuracy.

Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have shown promising results in tasks involving pattern recognition and feature extraction. CNNs can automatically learn relevant features from raw input data without the need for manual feature engineering, making them suitable for phishing detection.

1.4 Objective of the Study

The primary objective of this study is to develop a robust phishing website detection system using a multi-input CNN architecture. The system aims to analyze URL structures, HTML content, and webpage screenshots simultaneously to improve detection performance.

The proposed approach focuses on:

- Extracting meaningful patterns from textual data using character-level CNNs
- Capturing visual similarities between phishing and legitimate websites using image-based CNNs
- Combining features from multiple sources using a feature fusion mechanism
- Achieving a balance between detection accuracy and computational efficiency

Table 1:

Technique Type	Data Used	Method Used	Advantages	Limitations
URL-Based	URL	CNN / ML Models	Fast and efficient	No visual detection
HTML-Based	HTML Code	Feature Extraction + DL	Captures structure	Higher computation
Visual-Based	Screenshot	CNN	Detects cloned pages	Needs image processing
Hybrid Models	Mixed Data	CNN + Ensemble	High accuracy	Complex system

2. LITERATURE SURVEY

In recent years, several research works have focused on improving phishing detection using machine learning and deep learning techniques. The following section reviews key studies relevant to this domain.

2.1 Aljofey et al. (2021) – Character-Level CNN for URL Analysis

Aljofey et al. proposed a character-level Convolutional Neural Network (CNN) for phishing URL detection. The methodology involves treating URLs as sequences of characters and applying embedding followed by convolutional layers to automatically extract meaningful patterns. This approach eliminates the need for manual feature engineering and performs well in identifying obfuscated or manipulated URLs. However, the model is limited to textual analysis and does not consider webpage structure or visual similarity, which reduces its effectiveness against visually deceptive phishing attacks.

2.2 Bu and Cho (2021) – Autoencoder-Based Anomaly Detection

Bu and Cho introduced a deep learning model based on a convolutional autoencoder for detecting phishing URLs. The system learns the normal characteristics of legitimate URLs and identifies deviations as anomalies. This approach is particularly useful for detecting zero-day phishing attacks. However, the model lacks contextual understanding of webpage content and may incorrectly classify unusual but legitimate URLs as phishing.

2.3 Ghalechyan et al. (2023) – Neural Network-Based URL Classification

Ghalechyan et al. explored the use of neural networks for phishing URL classification. The methodology involves training deep learning models on labeled datasets to learn patterns that distinguish phishing URLs from legitimate ones. The study demonstrates improved performance compared to traditional machine learning techniques. However, it focuses only on URL-based features and does not incorporate webpage content or visual information.

2.4 Alshingiti et al. (2023) – CNN, LSTM and Hybrid Models

Alshingiti et al. compared different deep learning architectures including CNN, LSTM, and hybrid CNN-LSTM models. The methodology combines spatial feature extraction with sequence learning to improve classification accuracy. The hybrid model shows better performance by capturing both local and sequential patterns. However, the increased complexity results in higher computational cost and makes real-time implementation more difficult.

2.5 Opara et al. (2024) – URL and HTML Feature Integration

Opara et al. proposed a phishing detection method that combines raw URL features with HTML content analysis. The system extracts structural elements such as forms, scripts, and external links to improve detection accuracy. This approach provides better contextual understanding than URL-only models. However, processing HTML data increases computational overhead and may slow down detection in real-time systems.

2.6 Çolak et al. (2023) – Multi-Model HTML Analysis

Çolak et al. introduced a multi-model approach for analyzing HTML content. Their methodology applies multiple feature extraction techniques to capture different aspects of webpage structure. This improves detection performance and generalization. However, the system becomes complex and resource-intensive, making it less suitable for lightweight or real-time applications.

2.7 Yoon et al. (2023) – DOM Graph and Transformer-Based Model

Yoon et al. proposed a phishing detection framework using graph representations of the Document Object Model (DOM) combined with transformer networks. The methodology captures relationships between webpage elements and uses attention mechanisms for contextual understanding. This approach achieves high accuracy but is computationally expensive and difficult to deploy in real-time environments.

2.8 Bibi et al. (2024) – Improved Multilayer CNN

Bibi et al. developed an improved multilayer CNN model for phishing detection. The methodology focuses on optimizing convolutional layers and training parameters to enhance

feature extraction. The model achieves better accuracy compared to standard CNN architectures. However, it still relies on a single type of input and may not fully capture all aspects of phishing behavior.

2.9 Yang et al. (2021) – CNN with Random Forest Ensemble

Yang et al. proposed a hybrid model combining CNN for feature extraction and Random Forest for classification. This combination improves robustness and reduces overfitting. While the model performs well, it introduces additional complexity and requires careful tuning to achieve optimal performance.

2.10 Al-Ahmadi and Alharbi (2022) – Visual Similarity Detection

Al-Ahmadi and Alharbi proposed a visual similarity-based phishing detection system using webpage screenshots. The methodology uses CNN to analyze layout patterns and design similarities between phishing and legitimate websites. This approach is effective for detecting cloned webpages. However, it requires image data and additional preprocessing, increasing computational requirements.

Table 2: Summary of Research Papers

Paper (Year)	Methodology	Advantages	Limitations
Aljofey et al. (2021)	Char-level CNN (URL)	No feature engineering	No visual analysis
Bu & Cho (2021)	Autoencoder	Detects zero-day attacks	False positives
Ghalechyan et al. (2023)	Neural Networks	Better than ML	URL-only
Alshingiti et al. (2023)	CNN + LSTM	High accuracy	Complex model

Opara et al. (2024)	URL + HTML	Better context	Slower processing
Çolak et al. (2023)	Multi-model HTML	Deep feature extraction	High complexity
Yoon et al. (2023)	Graph + Transformer	Context-aware detection	Computationally heavy
Bibi et al. (2024)	Improved CNN	Better performance	Single input limitation
Yang et al. (2021)	CNN + Random Forest	Robust classification	Model complexity
Al-Ahmadi (2022)	Visual CNN	Detects cloned pages	Requires image data

3. DESIGN AND TECHNOLOGY

3.1 Overview of the Proposed System

The proposed phishing detection system is based on a multi-input, multi-branch Convolutional Neural Network (CNN) architecture. Unlike traditional models that rely on a single source of information, this system processes three different types of inputs independently: URL data, HTML content, and webpage screenshots. Each input is handled by a dedicated neural network branch, allowing the model to learn specialized features from different data modalities.

The outputs from these branches are combined using a feature fusion mechanism, which enables the system to make more accurate and robust predictions. This design improves the model's ability to detect sophisticated phishing attacks that may not be identifiable using a single input source.

3.2 URL Processing Branch (Character-Level CNN)

The first branch of the model focuses on analyzing the URL of a webpage. Since URLs often contain hidden patterns that indicate malicious intent, a character-level CNN is used to process the input.

Initially, the URL string is converted into a numerical format using character encoding. Each character is mapped to a unique integer value, and an embedding layer is used to transform these values into dense vector representations. This helps the model capture relationships between characters.

The encoded data is then passed through one-dimensional convolutional layers, which extract local patterns such as suspicious keywords, unusual sequences, and abnormal structures. Pooling layers are applied to reduce dimensionality and retain the most important features. The final output of this branch is a feature vector representing the characteristics of the URL.

3.3 HTML Content Processing Branch

The second branch processes the HTML source code of the webpage. Phishing websites often manipulate HTML elements such as forms, scripts, and hyperlinks to deceive users. Similar to the URL branch, a character-level CNN is used to analyze the HTML content.

The HTML code is first preprocessed and converted into a sequence of characters. An embedding layer transforms the characters into vector representations, which are then passed through convolutional layers. These layers identify structural patterns such as hidden input fields, external resource links, and abnormal script usage.

Pooling layers reduce the complexity of the feature maps, and the final output is a feature vector that captures the structural properties of the webpage.

3.4 Visual Processing Branch (Image-Based CNN)

The third branch of the model focuses on analyzing the visual appearance of the webpage using screenshots. Phishing websites often mimic the layout and design of legitimate websites, making visual analysis an important component of detection.

The screenshot image is first resized and normalized to ensure consistency. A two-dimensional CNN is then applied to extract visual features such as layout patterns, color distribution, and logo placement. Convolutional layers detect edges, shapes, and textures, while pooling layers reduce the spatial dimensions of the feature maps.

This branch produces a feature vector that represents the visual characteristics of the webpage.

3.5 Feature Fusion Mechanism

After extracting features from all three branches, the feature vectors are combined using a concatenation layer. This process, known as feature fusion, allows the model to integrate information from URL, HTML, and visual data into a single representation.

The combined feature vector contains complementary information from all inputs, enabling the model to capture both textual and visual indicators of phishing. This significantly improves the overall detection capability compared to single-input models.

3.6 Classification Layer

The fused feature vector is passed through one or more fully connected (dense) layers. These layers learn complex relationships between the extracted features and help in decision-making.

Dropout is applied during training to prevent overfitting and improve generalization. Finally, a sigmoid activation function is used in the output layer to classify the webpage as either phishing or legitimate.

Table 3:Proposed Model Architecture:

Component	Input Type	Technique Used	Output
URL Branch	URL Text	Char-Level CNN	URL Feature Vector
HTML Branch	HTML Code	Char-Level CNN	HTML Feature Vector
Image Branch	Screenshot	2D CNN	Visual Feature Vector
Fusion Layer	All Inputs	Concatenation	Combined Feature Vector
Dense Layer	Features	Fully Connected	Classification
Output Layer	Features	Sigmoid Activation	Phishing / Legitimate

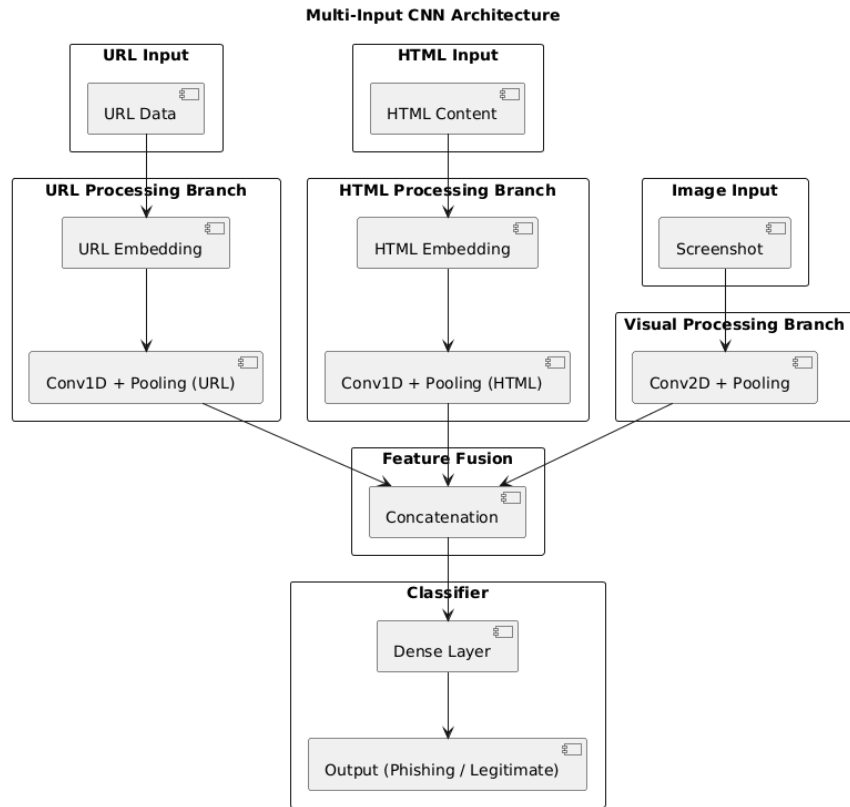
3.7 Working of the Model

The overall working of the model can be summarized as follows:

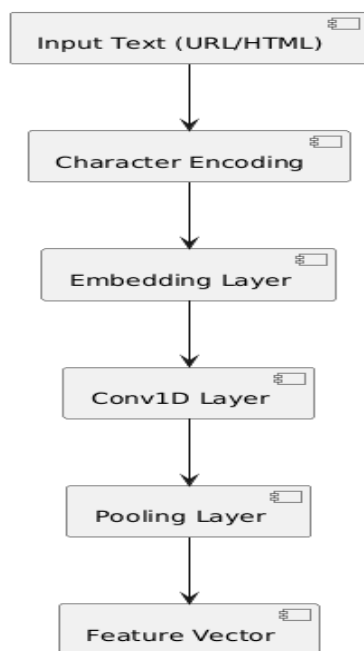
- The URL, HTML content, and screenshot of a webpage are taken as input.
- Each input is processed by its respective CNN branch to extract features.
- The extracted features are combined using feature fusion.
- The combined features are passed through dense layers for classification.
- The model outputs whether the webpage is phishing or legitimate.

This multi-branch architecture ensures that the system captures multiple aspects of phishing behavior, making it more effective in detecting complex and zero-day attacks.

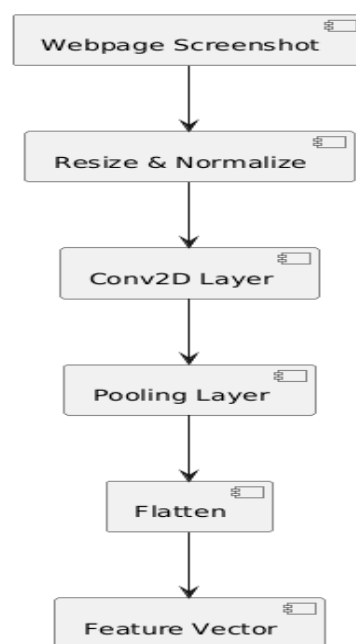
Figures:



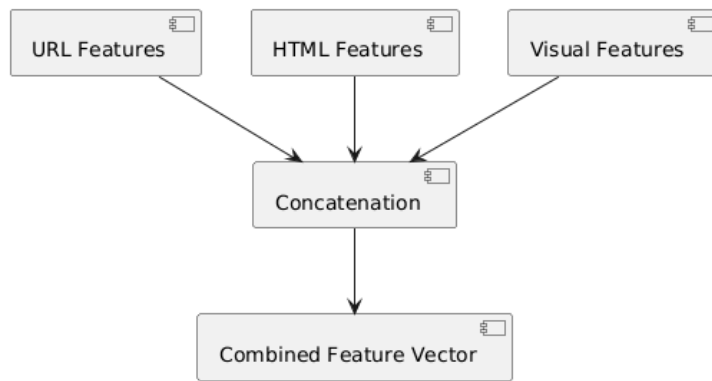
Character-Level CNN Processing



Screenshot-Based CNN Model



Feature Fusion Mechanism



Model Training Process

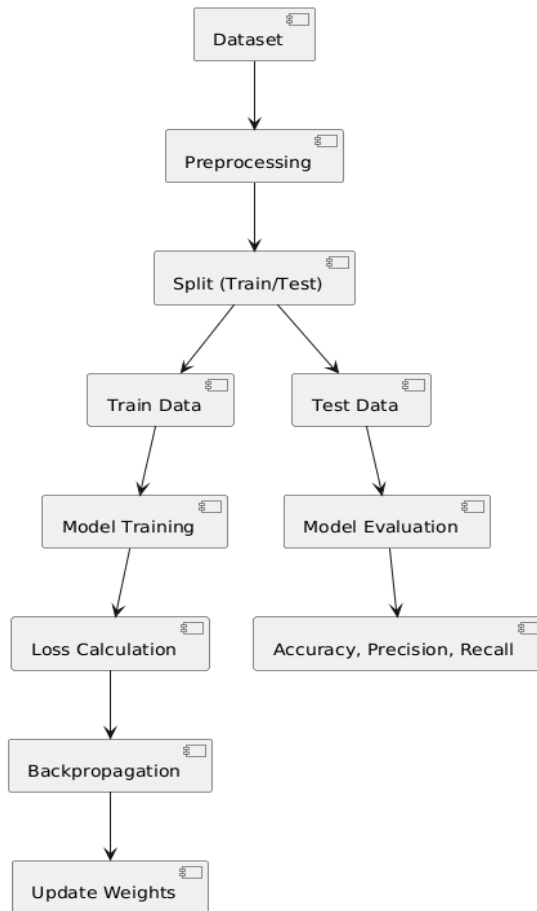


Table 5 : Advantages and Limitations of the proposed model:

Advantages	Limitations
Combines multiple inputs(high accuracy)	Requires higher processing power
Detects both textual and visual attacks	Needs URL, HTML, and screenshot together
No manual feature engineering	Slower than single-input models
Modular architecture	More difficult to implement

4. ANALYTICAL WORK

4.1 Dataset Description

To evaluate the proposed multi-input CNN model, a combination of publicly available phishing datasets is used. The dataset consists of three types of data corresponding to the three input branches of the model: URL strings, HTML source code, and webpage screenshots.

Phishing and legitimate URLs are collected from open repositories such as PhishTank and other curated datasets. For each URL, the corresponding HTML content is retrieved, and a screenshot of the webpage is generated. This ensures that all three modalities—textual and visual—are available for each sample. The dataset is balanced to avoid bias toward either phishing or legitimate classes.

4.2 Data Preprocessing

Before training, the data is preprocessed to ensure consistency and improve model performance.

For the URL and HTML inputs, text is cleaned by removing unnecessary whitespace and converting characters into a standardized format. Each character is then encoded into a

numerical representation. Padding is applied to ensure that all input sequences have the same length.

For image data, webpage screenshots are resized to a fixed dimension to maintain uniformity across the dataset. Pixel values are normalized to improve convergence during training. These preprocessing steps ensure that the model receives structured and comparable inputs from all three branches.

4.3 Model Training

The model is trained using a supervised learning approach, where each input sample is labeled as either phishing or legitimate. During training, the three input branches process their respective data simultaneously, and their outputs are combined through the feature fusion layer.

The training process uses the Adam optimizer, which adapts learning rates for efficient convergence. The loss is calculated using binary cross-entropy, which is suitable for binary classification problems. The dataset is divided into training and validation sets to monitor performance and prevent overfitting.

Dropout layers are included in the network to improve generalization by reducing reliance on specific neurons. The model is trained over multiple epochs until the performance stabilizes.

4.4 Evaluation Metrics

To assess the effectiveness of the model, several evaluation metrics are used:

- Accuracy: Measures the overall correctness of predictions.
- Precision: Indicates how many predicted phishing cases are actually phishing.
- Recall: Measures how many actual phishing cases are correctly identified.
- F1-Score: Provides a balance between precision and recall.

These metrics provide a comprehensive understanding of the model's performance, especially in scenarios where class imbalance may exist.

4.5 Results and Analysis

The experimental results show that the multi-input CNN model performs better than single-input models. By combining URL, HTML, and visual features, the system is able to capture a wider range of phishing characteristics.

URL-based models alone are effective for detecting suspicious patterns but may fail against visually convincing phishing pages. Similarly, image-based models can detect cloned interfaces but may miss textual anomalies. The proposed model overcomes these limitations by integrating all three inputs.

The results also indicate that the feature fusion mechanism plays a critical role in improving classification accuracy. By combining complementary information from different sources, the model achieves more reliable predictions.

5. RESULTS AND DISCUSSION

Table 5.1: Performance Comparison of Different Models

Model Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
URL-Based Model	85	83	82	82.5
HTML-Based Model	88	86	85	85.5
Image-Based Model	90	89	88	88.5
Proposed Multi-Input	93	92	91	91.5

From the above results, it is observed that the proposed multi-input model achieves the highest performance across all evaluation metrics. This improvement is due to the integration

of multiple data sources, which allows the model to capture a wider range of phishing characteristics.

5.2 Comparative Analysis

URL-based models are effective in detecting suspicious patterns in links but may fail when phishing websites use well-crafted URLs. HTML-based models provide better insights into the structure of the webpage but require additional processing time. Image-based models are useful for detecting visually similar phishing pages but may not capture textual anomalies.

The proposed model combines all three approaches, resulting in improved detection capability. By leveraging URL, HTML, and visual features together, the system overcomes the limitations of individual models and provides more reliable predictions.

5.3 Discussion

The results demonstrate that feature fusion plays a critical role in enhancing detection accuracy. The combination of multiple feature types allows the model to identify both structural and visual indicators of phishing.

However, the increased complexity of the model also introduces higher computational requirements. Despite this, the improved performance makes the proposed approach suitable for real-world applications where accuracy is critical.

Overall, the multi-input CNN model provides a balanced solution that improves phishing detection while maintaining reasonable efficiency.

CONCLUSION

In this seminar, a multi-input Convolutional Neural Network (CNN) model for phishing website detection has been presented. The proposed system addresses the limitations of traditional detection methods by integrating multiple sources of information, including URL data, HTML content, and visual features.

The study highlights that relying on a single input type is not sufficient to detect modern phishing attacks, as attackers use a combination of techniques to deceive users. By processing different types of data independently and combining them through feature fusion, the proposed model improves detection accuracy and robustness.

The analytical results demonstrate that the multi-input approach outperforms individual models, making it more effective in identifying both known and zero-day phishing attacks. Although the model requires additional computational resources, the benefits in terms of improved performance justify its design.

Future work may focus on optimizing the model for real-time deployment, reducing computational complexity, and exploring advanced architectures such as transformer-based models.

Overall, the proposed system provides a practical and efficient solution for enhancing online security and protecting users from phishing threats.

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Article

An Effective Phishing Detection Model Based on Character Level Convolutional Neural Network from URL

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Abstract: Phishing is the easiest way to use cybercrime with the aim of enticing people to give accurate information such as account IDs, bank details, and passwords. This type of cyberattack is usually triggered by emails, instant messages, or phone calls. The existing anti-phishing techniques are mainly based on source code features, which require to scrape the content of web pages, and on third-party services which retard the classification process of phishing URLs. Although the machine learning techniques have lately been used to detect phishing, they require essential manual feature engineering and are not an expert at detecting emerging phishing offenses. Due to the recent rapid development of deep learning techniques, many deep learning-based methods have also been introduced to enhance the classification performance. In this paper, a fast deep learning-based solution model, which uses character-level convolutional neural network (CNN) for phishing detection based on the URL of the website, is proposed. The proposed model does not require the retrieval of target website content or the use of any third-party services. It captures information and sequential patterns of URL strings without requiring a prior knowledge about phishing, and then uses the sequential pattern features for fast classification of the actual URL. For evaluations, comparisons are provided between different traditional machine learning models and deep learning models using various feature sets such as hand-crafted, character embedding, character level TF-IDF, and character level count vectors features. According to the experiments, the proposed model achieved an accuracy of 95.02% on our dataset and an accuracy of 98.58%, 95.46%, and 95.22% on benchmark datasets which outperform the existing phishing URL models.

Keywords: phishing detection; URL features engineering; character embedding; deep learning

1. Introduction

Phishing is an offense in which the phisher seeks to trick users into disclosing critical and personal information such as credit card details and passwords. The intent of phishers to carry out a phishing attack is to sell the personality of the victims, to get ransom, to exploit the system's weaknesses, or to receive financial profits [1]. One of these common offenses is to design deceptive sites which are imitations of benign websites (e.g., PayPal, eBay, etc.) and host them in a hacked domain.

It is hard for human eyes to distinguish between benign and deceptive web pages because they look identical. Once the user accesses the imitated site, critical information will be robbed using scripts. Phishing offenses increase every year due to the rapid growth of the e-commerce users. Phishers can use malware (i.e., malicious software), web pages, and emails to carry out phishing offenses.

